



COBIT 2019 for IT Governance Evaluation: A Cross-Sector Systematic Review

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ABSTRACT

IT governance is important because it helps make sure that a company's technology resources work well with its business goals and last for a long time. This study does a thorough review of existing research (called a systematic literature review) to look at how the COBIT 2019 framework is being used in different industries and areas of governance. All the chosen studies were looked at and sorted into groups based on the type of organization, the COBIT 2019 areas, and the goals of governance. The results show that COBIT 2019 is used in many different areas, like education, government, healthcare, industry, and companies that operate in the digital marketplace. The most common areas that come up are Align, Plan and Organize (APO), Build, Acquire and Implement (BAI), Deliver, Service and Support (DSS), and Evaluate, Direct and Monitor (EDM). Among these, the DSS and APO domains are most common as generally.

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1. INTRODUCTION

Governance designers can strategically blend control, coordination, incentives, and trust across three distinct layers: analog, augmented, and automated. By integrating these mechanisms, organizations can develop "hybrid configurations" that allow for a smooth, modular transition toward modernization [1]. After establishing the concept of organizational AI governance, we position it within the broader governance framework of the organization, defined as a set of organizational practices. We specifically highlight its interconnections with three primary governance domains: corporate governance, IT governance, and data governance [2]. The increasing complexity of IT environments, driven by cloud computing, digital transformation initiatives, and heightened cybersecurity threats, has intensified the need for structured IT governance evaluation. Organizations require systematic approaches to determine whether their IT governance practices effectively support reliable and sustainable IT services. Variations in organizational context and sector-specific requirements have led to diverse governance evaluation practices, which complicate the identification of consistent patterns and best practices across sectors.

Numerous critical factors have been identified as contributing to effective IT governance, including top management support, internal management mechanisms, organizational structure, staffing, and the strategic alignment of IT with business objectives. A fundamental aspect of IT governance is the distinction between governing bodies and management. Governing bodies are tasked with defining the mission, establishing policies, setting strategic direction, appointing and supervising IT management, exercising judgment and oversight, and monitoring performance. In contrast, management is responsible for executing operational activities [3].

In the domain of company growth and development, it is vital to understand the core concepts and principles of Information Technology, as this significantly contributes to enhancing competitiveness and driving progress. Organizations that not only grasp but also effectively implement Information Technology (IT) often experience improved management of their operational systems within a structured framework of IT governance [4]. Transformative governance must be adaptive to reflect the inherent complexity of environmental change. Adaptive governance Transformative governance needs to be adaptable to address the complexities of environmental change effectively. Adaptive governance is a process designed to enhance resilience by utilizing continuous opportunities for iterative learning. This approach involves adjusting responses to uncertainty, social conflicts, and complexity as situations evolve over time. Key principles include flexibility, collaboration, and a focus on ongoing improvement. is characterized as a process to enhance resilience that uses continuous opportunities for iterative learning, adjusting responses to uncertainty, social conflicts, and complexity over time [5].

By adopting the theoretical approach of COBIT 2019, this study provides practical recommendations to strengthen internal IT governance and ensure that the proposed strategies not only support but also align with strategic objectives and optimal operational sustainability [6]. As the information technology landscape becomes more complex, the need for a structured approach to managing IT risks is a key focus for service industry organizations. Effective IT Risk Management implementation requires a framework that is not only reliable but also adaptable to the dynamics and complexities involved. Frameworks such as COBIT have become a popular choice for service industry organizations. CoBIT (Control Objectives for Information and Related Technology) is a widely recognized IT management framework that offers a comprehensive framework for managing and overseeing enterprise IT processes. COBIT is an IT management and governance framework developed by ISACA (Information Systems Audit and Control Association) [7].

Research on IT governance and the increasing adoption of COBIT 2019 as an evaluation framework remains fragmented and largely sector-specific. Most prior Research focuses on individual organizational contexts or specific governance objectives, resulting in limited cross-sectoral insights into how COBIT 2019 is applied for IT governance evaluation. Despite the growing body of research on IT governance and the increasing adoption of COBIT 2019 as an evaluation framework, existing studies remain fragmented and primarily focused on specific sectors. Most prior Research examines individual organizational contexts or specific governance objectives, leading to limited cross-sector insights into how COBIT 2019 is applied for IT governance evaluation. As a result, there is a lack of a comprehensive understanding of common evaluation approaches, dominant COBIT domains, and recurring challenges across sectors.

This study focuses on using the Systematic Literature Review (SLR) which has the aim of analyzing IT governance at companies in Indonesia that use Cobit 2019. This study analyzes IT governance in Indonesian companies using Cobit 2019 through a Systematic Literature Review (SLR). A Systematic Literature Review (SLR) assesses and interprets all available research in a specific research area or phenomenon of interest, strengthening the foundation of study in that field. is used to assess and interpret the all-available research in a particular research area or phenomenon of interest where literature review can strengthen the foundation of study in a field of interest [8].

2. METHOD

The data collection for this study was carried out following the The management review protocol is widely used to explore, discover, and develop Research areas by providing researchers with the flexibility to modify SLR methods throughout the course of the study.



Figure 1. Systematic Literature Review Process

2.1 Research Questions

The research questions formulated in this study are as follows:

- RQ1: In which sectors has COBIT 2019 been applied for IT governance evaluation?
- RQ2: What COBIT 2019 domains are most frequently used in IT governance evaluation studies?
- RQ3: What evaluation approaches are commonly applied in IT governance studies using COBIT 2019?
- RQ4: What research gaps and challenges are identified in existing studies?

2.2 Data Sources and Search Strategy

Relevant studies were collected from academic databases such as Google Scholar and DOAJ. The search process was conducted using keywords including “IT governance”, “COBIT 2019”, “IT governance evaluation”, and “IT services”.

2.3 Inclusion and Exclusion Criteria

The inclusion criteria applied in this study are:

- Articles discussing IT governance evaluation using COBIT 2019
- Peer-reviewed journal or conference papers
- Publications from 2019 to 2024
- Articles written in English

The exclusion criteria include:

- Non-academic publications
- Studies not related to COBIT 2019
- Duplicate articles

2.4 Data Extraction and Analysis

Data extracted from the selected studies include publication year, research sector, applied COBIT 2019 domains, evaluation approaches, and key findings. The extracted data were analyzed descriptively to identify patterns, trends, and research gaps across sectors.

3. RESULTS AND DISCUSSION

The following procedure outlines the presentation of results using the COBIT 2019 framework. Relevant studies were sourced from Google Scholar and DOI-indexed journals, including those indexed at a minimum of SINTA 4 and Scopus. The overall research flow can be see with figure 2.

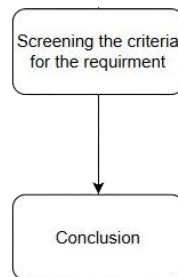


Figure 2. flow of COBIT 2019 Framework

3.1 Overview Selected Studies

The systematic search identified XX studies published between 2021 and 2026. The majority of these studies appeared in peer-reviewed journals indexed in SINTA and Scopus, which reflects increasing academic interest in COBIT 2019–based IT governance evaluation can be identified by table 1.

Tabel 1. Cobit 2019 Selected Studies.

Overview Selected Studies					
No	Author	Title	Index	Citation result	Cross Sector
1	Utomo, Diana Wijaya et all.	Leveraging COBIT 2019 to Implement IT Governance in SME Context: A Case Study of Higher Education in Campus A	Sinta 1	UTOMO, Diana, et al. Leveraging COBIT 2019 to Implement IT Governance in SME Context: A Case Study of Higher Education in Campus A. <i>CommIT (Communication and Information Technology) Journal</i> , 2022, 16.2: 129-141.	Education
2	Ade Winarni et all.	Audit Sistem Informasi Menggunakan COBIT 2019 (Studi Kasus SISFO Politeknik Enjinering Indorama) Ade	Sinta 4	Winarni, Ade, et al. "Audit sistem informasi menggunakan COBIT 2019 (studi kasus SISFO	Education

					Politeknik Enjineri Indorama)." <i>INFORMATION SYSTEM FOR EDUCATORS</i>	
3	Chandra Wijaya et all.	Audit Tata Kelola TI Menggunakan COBIT 2019 Domain APO-12 Pada Universitas Mikroskil	Sinta 4	Wijaya, Chandra, et al. "Audit tata kelola ti menggunakan cobit 2019 domain apo-12 pada universitas mikroskil." <i>Jurnal SIFO Mikroskil</i> 24.2 (2023): 197-210.	Education	
4	Satria et all.	Audit Tata Kelola Academic Information System Menggunakan Framework Cobit 2019	Sinta 2	Putra, Satriya Dwi, Herman Herman, and Anton Yudhana. "Audit tata kelola academic information system menggunakan framework cobit 2019." <i>Jurnal Teknologi Informasi dan Ilmu Komputer</i> 10.3 (2023): 467-474.	IT Sectoral	
5	U Alfianto et all.	Audit Tata Kelola Teknologi Informasi Menggunakan Framework Cobit 2019 Terhadap Pencapaian It Master Plan Lembaga Pelatihan Xyz	Sinta 4	Alfianto, Ufan, Irman Hermadi, and Sri Wahjuni. "Audit Tata Kelola Teknologi Informasi Menggunakan Framework COBIT 2019 Terhadap Pencapaian IT Master Plan Lembaga Pelatihan XYZ." <i>Syntax Literate; Jurnal Ilmiah Indonesia</i> 7.11 (2022): 16293-16307.	IT Sectoral	
6	Sumardiono et all.	Unveiling IT Governance Effectiveness in e-Puskesmas Using COBIT 2019	Sinta 2	Sumardiono, et al. "Unveiling IT Governance Effectiveness in e-Puskesmas Using COBIT 2019." <i>Scientific Journal of Informatics</i> 13.1 (2026).	Healthcare	
7	Adila S et all.	Identifikasi Level Pengelolaan Tata Kelola SIPERUMKIM Kota Salatiga berdasarkan COBIT 2019	Scopus	Safitri, Adila, Imam Syafii, and Kusworo Adi. "Identifikasi level pengelolaan tata kelola SIPERUMKIM kota Salatiga berdasarkan COBIT 2019." <i>Jurnal RESTI (Rekayasa Sistem Dan Teknologi Informasi)</i> 5.3 (2021): 429-438.	Gov Sectoral	
8	I Nyoman et all.	Evaluasi Tata Kelola Teknologi Informasi Di Dinas Pertanian Gianyar Menggunakan Cobit 2019	Sinta	Kesuma, I. N. R. W., Irman Hermadi, and Yani Nurhadryani. "Evaluasi Tata Kelola Teknologi Informasi di Dinas Pertanian Gianyar Menggunakan COBIT 2019." <i>Jurnal Teknologi Informasi dan Ilmu Komputer</i> (	Gov Sectoral	
9	Herliani, Meidina et all.	Evaluation Of It Governance Using Cobit 2019 On Regional Asset Management Agency Of Dki Jakarta	Sinta 5	Herliani, Meidina, and Wella Wella. "EVALUATION OF IT GOVERNANCE USING COBIT 2019 ON REGIONAL ASSET MANAGEMENT AGENCY OF DKI JAKARTA." <i>Jurnal Pilar Nusa Mandiri</i> 20.2 (2024): 176-182.	Healthcare	
10	Widasari et all.	Capability Level Assessment of IT Governance in the SIAP KOJA Application Using the COBIT 2019 Framework	Sinta 3	Widasari, Yesya Najwa, and Nabila Rizky Oktadini. "Capability Level Assessment of IT	Gov Sectoral	

				Governance in the SIAP KOJA Application Using the COBIT 2019 Framework." <i>Journal of Applied Informatics and Computing</i> 9.6 (2025): 3706-3715.	
11	Faza ahmad et all.	COBIT 2019 Implementation for Enhancing IT Governance in Educational Institutions	Sinta 3	Ahmad Faza. "COBIT 2019 implementation for enhancing IT governance in educational institutions." <i>JISKA (Jurnal informatika sunan kalijaga)</i> 8.3 (2023)	Education
12	Morris William Tangka, et all.	Information Technology Governance Analysis Using The COBIT 2019 Framework at XYZ Institution	Sinta 2	Tangka, George Morris William, Andrew Tanny Liem, and Joe Yuan Mambu. "Information Technology Governance Audit Using the COBIT 5 Framework at XYZ University." <i>2020 2nd International Conference on Cybernetics and Intelligent System (ICORIS)</i> . IEEE	IT Sectoral
13	Merryana Lestari et all.	Information Technology Governance Design in DevOps-Based E-Marketplace Companies Using COBIT 2019 Framework	Scopus	Lestari, Merryana, Ade Iriani, and Hendry Hendry. "Information Technology governance design in DevOps-based E-marketplace companies using COBIT 2019 framework." <i>INTENSIF: Jurnal Ilmiah Penelitian dan Penerapan Teknologi Sistem Informasi</i> 6.2 (2022): 233-252.	IT Sectoral
14	Utama, Danu Prawira Muhammad et all.	Audit Manajemen Masalah Teknologi Informasi Menggunakan Kerangka Kerja Cobit 2019 Domain Dss03	Sinta 3	Utama, Danu Prawira, Alva Hendi Muhammad, and Agus Purwanto. "Audit Manajemen Masalah Teknologi Informasi Menggunakan Kerangka Kerja Cobit 2019 Domain Dss03." <i>JIP1 (Jurnal Ilmiah Penelitian dan Pembelajaran Informatika)</i> 8.3 (2023): 839-846.	IT Sectoral

The following table shows several ways to obtain the required data values, including using journal indexes in Sinta and Scopus as baseline comparisons for the selected studies. This data was searched using the Google Scholar index and scientific articles from 2021 to present, using the primary keyword "cobit 2019."

3.2 Sector Distribution of COBIT 2019 Studies

Table 2 explains the distribution of sectors used to classify questions in the form of R1 regarding the SLR methodology used in this study. This methodology is key to applying a division classification that better aligns with existing guidelines and improves clustering performance.

Tabel 2. Data Extraction

Overview Selected Studies			
No	Author - Year	sector	Domain
1	Utomo, Diana Wijaya et all. (2022)	Higher Education SME	EDM1, APO6, BAI8, DSS6
2	Ade Winarni et all. (2023)	Higher Education (SME)	EDM1, APO6, BAI8, DSS6

3	Chandra Wijaya et al. (2023)	Higher Education (SME)	APO1, APO12
4	Satria et al. (2023)	Higher Education (SME)	EDM1, APO1
5	U Alfianto et al. (2022)	Education / Training Institute	EDM03, APO11, BAI06, DSS03
6	Sumardiono et al. (2026)	Healthcare (Public Health Center)	APO12, DSS02
7	Adila S et al. (2021)	Government (Public Housing Agency)	APO12, DSS02, DSS03
8	I Nyoman et al. (2022)	Government (Agriculture Office)	MEA03, BAI04, EDM03
9	Herliani, Meidina et al. (2024)	Government (Regional Asset Management Agency)	EDM05, APO08, APO12
10	Widasari et al. (2025)	Government	APO11, BAI05
11	Faza ahmad et al. (2023)	Education	BAI04, BAI05, BAI11
12	Morris William Tangka, et al. (2022)	Education (Higher Education Institution)	BAI03, BAI07, BAI10, DSS01, DSS05
13	Merryana Lestari et al. (2022)	Industry	APO03, APO04, BAI04, BAI06, BAI11, DSS03
14	Utama, Danu Prawira Muhammad et al. (2023)	Industry	DSS03

The distribution of the selected COBIT 2019 studies across different sectors was looked at based on the organizational settings mentioned in each article, as shown in Table 2. The findings show that the majority of the studies were from the education sector (7 studies), and the next highest was government organizations (4 studies). In comparison, only a small number of studies were found in the industrial sector ($n = 2$) and the healthcare sector ($n = 1$).

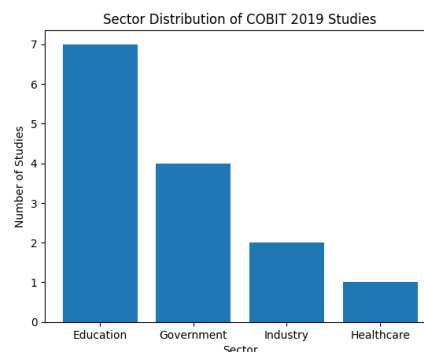


Figure 3. Sector Distribution of COBIT 2019

This shows that COBIT 2019 is mostly used by schools and universities, as well as training groups. There aren't many studies on how it's used in hospitals or factories, which means people aren't looking much into how it works in those areas yet. This gap offers chances for more research, especially in healthcare information systems and industrial IT governance.

3.3 COBIT 2019 Domain Usage

An analysis of the COBIT 2019 domains utilized in the selected studies was conducted to determine the most frequently applied governance areas. According to the domain mapping in Table 2, the APO (Align, Plan, and Organize) domain emerged as the most frequently used, appearing in 10 studies. The BAI (Build, Acquire, and Implement) and DSS (Deliver, Service, and Support) domains were each identified in eight studies, while the EDM (Evaluate, Direct, and Monitor) domain was present in six studies. In contrast, the MEA (Monitor, Evaluate, and Assess) domain was identified only once. It appeared only once.

3.4 Evaluation Value

Evaluation approaches in COBIT 2019 studies were examined to determine prevalent assessment methods in IT governance research. Audit-based evaluation emerged as the most common approach, identified in six studies.

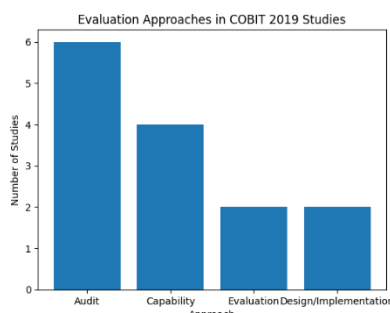


Figure 4. Evaluation Distribution of COBIT 2019

Figure 4 shows the distribution of evaluation approaches used in COBIT 2019 studies. Audit-based approaches dominate the selected literature with six studies, followed by capability level assessment (four studies).

3.5 Research Gaps

Current studies on COBIT 2019 show that there's a noticeable lack of balance when it comes to which industries are being looked at and how the framework is being used in different areas. While the education and government areas are often talked about, healthcare and industrial settings are usually ignored. This shows there is a big gap in our knowledge about how to adapt COBIT 2019 to fit the specific rules and work needs of medical information systems and industrial IT. The focus is spread unevenly across all the areas in the framework. Most writings focus on the APO (Align, Plan, and Organize), BAI (Build, Acquire, and Implement), and DSS (Deliver, Service, and Support) areas. On the other hand, the MEA (Monitor, Evaluate, and Assess) area is almost completely overlooked. This shows that the academic community focuses more on the steps needed to set up and carry out research, rather than on the ongoing checking and making sure that things work well over time, which is important for good, lasting governance.

4. CONCLUSION

The results show that the most commonly used areas in COBIT 2019 are Align, Plan, and Organize (APO); Build, Acquire, and Implement (BAI); Deliver, Service, and Support (DSS); and Evaluate, Direct, and Monitor (EDM). Among these, the DSS and APO domains are the most common, especially in studies related to IT service management, risk management, and problem management. Several studies also focus on specific goals, such as DSS03 (Managed Problems), APO12 (Managed Risk), and BAI04/BAI06 (availability and change management), showing that companies typically pay more attention to operational control and reliable services. The way COBIT domains are organized depends on the type of organization and the governance area they are focusing on. The SLR finds that COBIT 2019 offers a thorough and flexible framework for assessing and establishing IT governance across industries. Future studies could build on this review by including analyses of capability levels and comparisons of maturity to better understand how effective governance is across industries.

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